

Creating and Using XR for Environmental Communication: Three Exploratory Case Studies

Barbara Buljat Raymond*, †Daniel Pimentel, ‡Kay Poh Gek Vasey

*Université Côte d'Azur, CNRS, GREDEG; Faculty of Economics and Business, University of Rijeka; Balkan Institute of Science and Innovation of the Université Côte d'Azur; † University of Oregon; ‡ Meshminds

ABSTRACT

Despite growing investment in environmental media campaigns, the link between awareness and action remains inadequate. One promising approach involves using extended reality (XR) technologies – augmented reality (AR) and virtual reality (VR) – to provide audiences with direct interactions with environmental dangers. The current work presents three interdisciplinary case studies highlighting how brands, non-governmental organizations, and academic institutions are leveraging XR for environmental communication.

Keywords: XR technologies, augmented reality (AR), virtual reality (VR), environmental communication, pro-environmental behavior (PEB).

1 INTRODUCTION

Studies continue to show an unprecedented degradation of the environment, an issue exacerbated by human activity [1]. Despite the magnitude of environmental threats, organizations struggle to find an effective way to motivate pro-environmental behaviour (PEB): the abstract nature of environmental data and the gap between the causes and consequences of environmental problems may be an obstacle for people to understand their seriousness [2]. Recently, attention from the industry and academia has been drawn to XR technologies such as augmented reality (AR) and virtual reality (VR), for making environmental communication more immersive, engaging, and convincing.

Global leaders in environmental communications, such as the United Nations Environment Programme (UNEP) and the World Wildlife Fund (WWF), have recognised this trend and have begun to incorporate immersive virtual experiences into their social media campaigns to motivate community movements (e.g. see the WWF's campaign "Take a Photo With the Leopard"). Besides public dialogue, institutions have also been experimenting with another mechanism for motivating PEB – body transfer (BT) in VR [3], to let people experience "living in someone else's shoes".

The scientific community has increasingly invested in studying the impact of AR/VR for addressing environmental sustainability. For example, immersive storytelling has been used to teach marine science [4], simulate the impacts of plastic pollution [5] and embody endangered animals [6]. However, the research in this area is still in its infancy. This paper contributes to the debate and provides an interdisciplinary investigation into the use of immersive technologies for addressing an urgent environmental problem: plastic pollution. In the subsequent sections we highlight three case studies wherein AR- and VR-based experiences were designed and distributed for the purpose of raising awareness of the problem and catalysing individual action. In doing so, we seek to

demonstrate the potential of AR and VR for advancing environmental conservation efforts.

2 CASE STUDIES

2.1 Endangered animals in AR

In this case study, we used a Design Science Research [7] approach and a focus group method to obtain end users' feedback about an AR system showing a sea animal endangered due to plastic pollution (Figure 1A). Participants (N=7) were master's students ($M_{age}=23.7$, $SD=0.76$) at one European faculty of economics. We analyzed the data inductively according to qualitative research guidelines that ensure scientific rigor [8].

First, seeing endangered animals in AR elicited emphatic response from some participants, which may result in an emotional trigger that drives PEB. Second, participants supported the concept of using AR to bring environmental issues "psychologically closer", by attaching them to familiar and personally relevant places (e.g. their favourite beach). Third, visual fidelity and graphical representation of virtual objects may play an important role in such interventions. Although participants reacted with shock ("Oh, poor thing!", "Almost like realistic", "Too good", "It's cool!"), they concluded that improved, more realistic visuals are necessary to be effective and generate stronger empathy. In conclusion, while our findings suggest that AR could improve traditional environmental communication, its design has to match users' mental expectations in order to effectively motivate PEB.

2.2 VR Embodiment of Endangered Animals

The goal of the second case study was to examine the feasibility and pro-environmental implications of embodying threatened wildlife in VR (animal embodiment). We conducted a mixed methods field study at a large marine science center in the Southeastern United States. Participants evaluated an educational VR simulation "Project SHELL", in which they embodied a threatened Loggerhead Sea turtle as it encounters various environmental threats (Figure 1B). First, three senior staff members provided qualitative feedback about the simulation. Then, we conducted a within-subjects field experiment (N=12, $M_{age}=15$, $SD=.95$) to assess participants' perceptions of the simulation, as well as, its effect on environmental attitudes. Body transfer was enabled by synchronizing users' movements with a virtual sea turtle, and by haptic feedback corresponding to in-game threats.

Broadly, the results suggest that, even with a short intervention, BT with threatened wildlife can be achieved through VR ($M_{BT}=3.71$, $SD=.52$), and can contribute to connectedness with nature (INS) (before ($M=4.25$, $SD=1.48$) after ($M=4.66$, $SD=1.3$), $t(11)=-1.82$, $p<.05$). Moreover, observational data was useful to get insight into post-pandemic VR usage in public spaces. As students were required to wear masks during the entirety of the camp, a common issue was screen fogging, inhibiting visibility. This may have impeded illusory BT considering that visual quality is an important antecedent to presence-related constructs [9]. Discussions with the science center's staff emphasized the fact that while accessing

*barbara.buljat-raymond@etu.univ-cotedazur.fr;
†pimend@uoregon.edu; ‡kay@meshminds.com

immersive learning content is important, it is all for nothing if staff do not have the capacity and resources to facilitate the experiences effectively.

2.3 AR Interactions with Environmental Dangers

A promising strategy to motivate PEB is to provide audiences with impacts of their consumption behaviors in real time. In the third case study, we created an AR experience to test the efficacy of this approach. Specifically, the SEA circular project and The MeshMinds Foundation, in collaboration with UNEP, launched the social media AR campaign “100 Days to #BeatPlasticPollution”. Four AR experiences and games (Figure 1C) were available as Instagram camera effects. We conducted a between-subjects experiment (2 groups: control and experimental) to investigate the extent to which such AR experiences could raise environmental awareness, intentions, and actions [10]. Two surveys were used to assess impact of a 5.5 week-long intervention over a 3 month period. Participants (N=40, aged 13-25, 23 female), were students from multiple universities.

There were no significant differences between the two groups on any of the measures assessed before or after the intervention (all p 's > .05), except the self-reported beliefs that individual actions can mitigate plastic pollution, that was higher in the intervention group $F(1, 32) = 7.75, p < .01$. Specifically, participants who interacted with the AR experiences reported significantly greater beliefs in their individual capacity to mitigate plastic pollution ($M=4.56, SD=.62$) compared to the control group ($M=3.88, SD=.81$). Environmental awareness increased for both groups, though the difference was only significant for participants in the experimental group ($F(1,30) = 6.57, p = .01$). Moreover, beliefs that individual actions influence plastic pollution significantly increased after the intervention for the experimental group, $F(1,30) = 5.95, p = .02$.

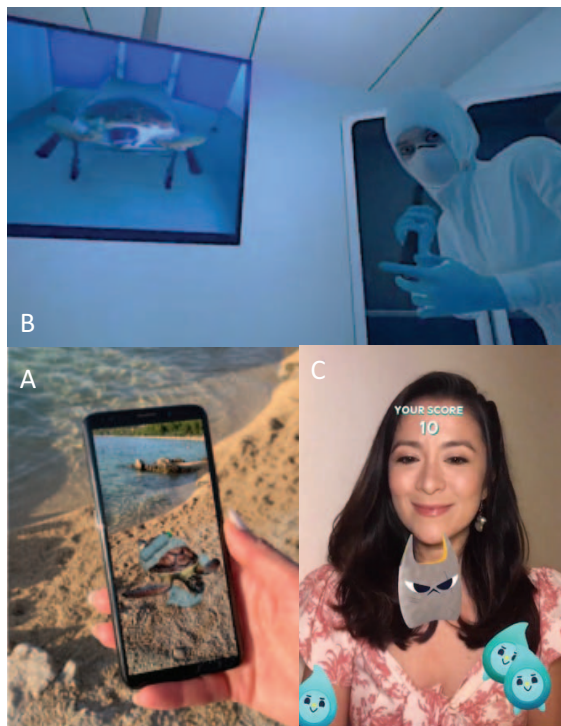


Figure 1: Screenshots of immersive experiences from the case studies 1(A), 2(B) and 3(C).

3 CONCLUDING REMARKS

Despite the urgency of the global environmental crisis, environmental communicators struggle to find an effective way to motivate pro-environmental behaviors. One way to make environmental communication more engaging and persuasive could involve XR technologies. Immersive AR and VR experiences can leverage affordances such as presence and embodiment to facilitate direct experiences that influence environmental awareness and behavior. Three distinct international studies tested the efficacy of XR-based environmental conservation strategies using varied methodological approaches. Collectively, the results suggest that AR and VR experiences that depict environmental threats, either onto the self or others, can contribute to short- and long-term pro-environmental outcomes, namely attitudes and intentions. This investigation presents modest contributions. First, the synthesis of three interdisciplinary case studies generates new knowledge about the design, development, distribution and impact of XR technologies for promoting individual and collective PEBs. Second, it provides concrete examples of how brands, organizations, and educational institutions are incorporating XR storytelling into their communication activities to engage their audiences, which could be useful for policy makers and communicators. We encourage scholars and practitioners to further explore the potential of XR applications for addressing important societal issues such as environmental sustainability, and spotlight the unique features of each XR platform which may differentially increase pro-environmental outcomes through varied mechanisms.

REFERENCES

- [1] IPCC, 2022: Summary for Policymakers. Cambridge University Press, Cambridge, UK and New York, NY, USA, pp. 3-33, doi:10.1017/9781009325844.001.
- [2] Van der Linden, S., Maibach, E., & Leiserowitz, A. (2015). Improving public engagement with climate change: Five “best practice” insights from psychological science. *Perspectives on Psychological Science*, 10, 758–763. doi: 10.1177/1745691615598516
- [3] Steuer, J. (1992). Defining Virtual Reality: Dimensions Determining Telepresence. *Journal of Communication*. <https://doi.org/10.1111/j.1460-2466.1992.tb00812.x>
- [4] Fauville, G., Queiroz, A. C. M., Hambrick, L., Brown, B. A., & Bailenson, J. N. (2021). Participatory research on using virtual reality to teach ocean acidification: a study in the marine education community. *Environmental Education Research*, 27(2), 254–278. <https://doi.org/10.1080/13504622.2020.1803797>
- [5] Buljat, B. (2022) "Augmented Reality For Nudging Green Behavior: Design, Evaluation And Implementation". ECIS 2022 Research Papers. 56. https://aiselaisnet.org/ecis2022_rp/56
- [6] Pimentel, D., Kalyanaraman, S. The effects of embodying wildlife in virtual reality on conservation behaviors. *Sci Rep* 12, 6439 (2022). <https://doi.org/10.1038/s41598-022-10268-y>
- [7] Hevner, A.R.; March, S.T.; and Park, J. (2004) Design research in information systems research. *MIS Quarterly*, 28, 75–105.
- [8] Gioia, D. A., Corley, K. G., & Hamilton, A. L. (2013). Seeking Qualitative Rigor in Inductive Research: Notes on the Gioia Methodology. *Organizational Research Methods*, 16(1), 15–31. <https://doi.org/10.1177/1094428112452151>
- [9] Zou, W., Yang, F., Zhang, W., Li, Y., & Yu, H. (2018). A Framework for Assessing Spatial Presence of Omnidirectional Video on Virtual Reality Device. *IEEE Access*, 6, 44676–44684. <https://doi.org/10.1109/ACCESS.2018.2864872>
- [10] Vasey, K., Bos, O., Nasser, F., Tan, A., Li JunTing, B., Tat Khoo, E., & Marsh, T. (2019). Water Bodies: VR Interactive Narrative and Gameplay for Social Impact. In *The 17th International Conference on Virtual-Reality Continuum and its Applications in Industry* (pp. 1-2).